llmXive Automated Review of Identifying stimulus-driven neural activity patterns in multi-patient intracranial recordings

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer

Verdict: major_revision_writing
Free-form review body

Strengths - **Comprehensive Scope**: The manuscript provides an exceptionally thorough overview of the challenges and methodologies involved in identifying stimulus-driven neural activity, particularly in the context of multi-patient intracranial recordings (iEEG). - **Methodological Breadth**: It effectively covers a wide range of analytical approaches, from basic GLMs and MVPA to advanced techniques like Hyperalignment, Gaussian Process models, and Hierarchical Topographic Factor Analysis (HTFA). - **Conceptual Clarity**: The text does an excellent job of explaining complex concepts (e.g., the difference between univariate and multivariate patterns, static vs. dynamic measures) with clear definitions and logical progression. - **Visual Aids**: The figure inventory suggests a rich set of visualizations (e.g., spatial vs. temporal resolution, electrode coverage, joint stimulus-activity models) that, once compiled, would significantly aid understanding.

Concerns - **Compilation Failure**: The LaTeX source currently fails to compile. The use of the 'llncs' class (Lecture Notes in Computer Science) and 'pxfonts' package suggests a specific formatting requirement that is not met in the current environment. Without a compiled PDF, the final layout and figure placement cannot be verified. - **Missing References**: The provided 'reference.bib' file only contains the arXiv entry for the paper itself. The text cites over 100 external sources (e.g., 'Jaco21', 'Herc09', 'HaxbEtal11'). These are missing from the bibliography file, which will cause compilation errors and render the citations invalid. - **Contextual Mismatch**: The title and text explicitly frame this

as ”Chapter 41” of a forthcoming book (”Intracranial EEG for Cognitive Neuroscience”). If the intent is to submit this as a standalone arXiv preprint or journal article, the ”Chapter” framing is inappropriate and needs to be removed. The abstract and introduction should be rewritten to stand alone. - **Figure Path Discrepancy**: The LaTeX code references figures in ‘figs/‘, but the inventory shows files in both ‘figs/‘ and ‘figs-sanitized/figs/‘. This inconsistency will prevent the figures from rendering in the compiled document. - **Lack of Original Data**: As a review chapter, this is acceptable, but the ”Results” or ”Case Studies” sections rely entirely on cited literature. If this were intended as an original research paper, it would lack the necessary empirical contribution. However, given the text, it appears to be a review, so this is a structural observation rather than a flaw, provided the genre is correctly identified.

Recommendation The paper is currently in a **draft state** and cannot be published or accepted in its current form. The primary issues are technical (LaTeX compilation, missing bibliography entries, incorrect figure paths) and structural (book chapter framing vs. standalone paper).

The verdict is **major_revision_writing**. The manuscript requires a significant rewrite to: 1. Correct the LaTeX source to ensure successful compilation (fixing class, packages, and figure paths). 2. Populate the bibliography with all cited references. 3. Reframe the document from a ”book chapter” to a ”review article” if that is the intended output, removing all book-specific metadata. 4. Once the text is stable and compiles, the content itself is strong and ready for peer review.

The Paper-Tasker should be re-run with a revision brief focused on ”Fixing LaTeX compilation errors, populating the bibliography, and re-framing the manuscript as a standalone review article.”

- [writing] Populate the bibliography file with all ~100 cited references currently missing from reference.bib to resolve compilation errors and validate citations.
- [writing] Correct LaTeX figure paths to match the actual file locations in the inventory (figs/ vs figs-sanitized/) and ensure the llncs class and pxfonts package are available or replaced.
- [writing] Rewrite the title, abstract, and introduction to remove ’Chapter 41’ and ’forthcoming in [Book Title]’ framing, converting the manuscript into a standalone review article.
- [writing] Remove draft markers such as ’Note: This chapter is forthcoming’ and ’Draft current as of [date]’ to finalize the manuscript status.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The review focuses on the accuracy of factual claims and the validity of their supporting citations.

The manuscript provides a comprehensive overview of methods for identifying stimulus-driven neural activity. However, several citations do not precisely support the specific claims made in the text, or the claims are stated more strongly than the cited evidence permits.

1. **Section 1.1 (Overview):** The text cites 'Jaco21' (Jacobs, 2021, "Color vision") to support the claim that "when a photopigment in a retinal photoreceptor absorbs light, this triggers a cascade of responses..." While the paper discusses color vision, it is an encyclopedia entry and likely does not provide the detailed mechanistic description of the phototransduction cascade implied. A primary source on phototransduction (e.g., Yau & Hardie, 2009) is more appropriate for this specific mechanistic claim.

2. **Section 1.2 (Activity):** The text cites 'Barr08' (Barrs, 2008, "The mystery and magic of glia") for the claim that glial cells play an important role in "synapse formation." While this review discusses glial functions, the specific claim about synapse formation is more rigorously established in primary literature (e.g., Ullian et al., 2001). The citation should be updated to a primary source or qualified to reflect the review nature of the current source.

3. **Section 2.1 (Single-channel signals):** The text cites 'MannEtal09a' (Manning et al., 2009) for the claim that "broadband shifts in power... can occur when the neurons in the underlying population change their firing rates." The cited paper demonstrates a *correlation* between broadband shifts and spiking. The text's phrasing ("can occur when") implies a causal or direct mechanistic link that the correlation study alone does not fully establish. The text should clarify that the source demonstrates a correlation, or cite a source that explicitly establishes the causal mechanism.

4. **Section 3.1 (GLMs):** The text cites 'NeldWedd72' (Nelder & Wedderburn, 1972) for the definition of GLMs. While this is the correct seminal paper, the text describes the link function producing "indicator vectors" (one-hot encoding) as a general feature of GLMs. This is a specific application (multinomial logistic regression) rather than the general definition of a GLM. The text should clarify that this is a specific instantiation for classification tasks.

5. **Section 3.5 (ISC):** The text cites 'HassEtal04' (Hasson et al., 2004) for ISC. The text states that ISC "treats the average signals recorded from other participants as a model of the stimulus dynamics." While this is a valid interpretation of the method's utility in naturalistic paradigms, the original paper focuses on "synchronization" and does not explicitly frame the average signal as a "model of the stimulus." The claim is a reasonable inference but should be qualified as an interpretation of the method's application rather than a direct claim from the cited paper.

These issues are primarily "writing" severity as they can be resolved by refining the text to more accurately reflect the scope and findings of the cited sources, or by updating the citations to more appropriate primary literature.

- [writing] Section 1.1: Citation 'Jaco21' (Color vision encyclopedia entry) does not support the specific mechanistic claim about the phototransduction cascade. Replace with a primary source on phototransduction.

- [writing] Section 1.2: Citation 'Barr08' is a review; the specific claim about glial synapse formation is better supported by primary literature (e.g., Ullian et al., 2001). Update citation or qualify the claim.
- [writing] Section 2.1: Citation 'MannEtal09a' shows correlation, but text implies causation ('can occur when'). Clarify that the source demonstrates correlation, not necessarily a direct causal mechanism.
- [writing] Section 3.1: Text describes GLM link functions producing 'indicator vectors' as a general feature, but this is specific to multinomial logistic regression. Clarify this is a specific instantiation for classification.
- [writing] Section 3.5: Text claims ISC treats average signals as a 'model of stimulus dynamics,' but the cited 'HassEtal04' focuses on synchronization. Qualify this as an interpretation of utility rather than a direct claim.

paper_reviewer_code_quality_paper

Verdict: minor_revision
The provided manuscript is a comprehensive survey chapter on identifying stimulus-driven neural activity patterns. However, from a code quality and reproducibility perspective, the LaTeX source exhibits significant hygiene issues that hinder its ability to be compiled and maintained as a standard scientific artifact.

First, the bibliography management is non-standard. The file ‘main.tex‘ contains a massive, manually formatted ‘thebibliography‘ environment with over 100 hardcoded entries. This approach is brittle, prone to typos, and impossible to update programmatically. The project includes a ‘reference.bib‘ file, yet the LaTeX source ignores it entirely. To ensure reproducibility and maintainability, the source must be refactored to use BibTeX or BibLaTeX, linking the ‘.tex‘ file to the ‘.bib‘ database via ‘\bibliographystyle‘ and ‘\bibliography{reference}‘ commands. This would allow for automatic sorting, deduplication, and formatting updates.

Second, there are inconsistencies in figure path resolution. The LaTeX source references figures using paths like ‘figs/spatial_vs_temporal_resolution‘, but the provided file list indicates that the primary figures are in ‘figs/‘ while sanitized versions exist in ‘figs-sanitized/figs/‘. If the build environment expects the sanitized versions or if the paths are relative to a different root, the compilation will fail with "file not found" errors. The code should either standardize on a single directory structure or include a build script (e.g., a Makefile or shell script) that copies or symlinks the necessary assets before compilation.

Finally, the use of the ‘pxfonts‘ package is a known anti-pattern in modern LaTeX. This package is deprecated and often causes conflicts with other packages (such as ‘natbib‘ or ‘hyperref‘) and can lead to missing glyphs or incorrect math spacing. Replacing ‘pxfonts‘ with the actively maintained ‘newtxtext‘ and ‘newtxmath‘ packages would resolve these potential compilation failures and ensure the document renders correctly across different LaTeX distributions.

While the content is high-quality, the lack of standard tooling (BibTeX) and the use of deprecated packages make the artifact difficult to reproduce from scratch without manual intervention.

- [writing] The manuscript references multiple figure files (e.g., figs/spatial_vs_temporal_resolution, figs/signals) that are not present in the provided LaTeX source or the compiled PDF list. While the sanitized versions exist, the primary source code relies on paths that may not resolve in a standard build environment without manual intervention. Ensure all figure paths in the LaTeX source match the actual repository structure or provide a build script to handle path resolution.
- [writing] The bibliography is manually formatted using a custom thebibliography environment with hardcoded entries rather than using BibTeX/BibLaTeX with the provided reference.bib file. This violates standard reproducibility practices for scientific papers, making it difficult to update citations, check for duplicates, or regenerate the bibliography. Refactor to use \bibliographystyle and \bibliography{reference} commands.
- [writing] The LaTeX source uses the 'pxfonts' package, which is known to have compatibility issues with modern LaTeX distributions and can cause font rendering errors or conflicts with other packages (e.g., natbib). Replace with a more robust font package like 'newtxtext' and 'newtxmath' to ensure consistent compilation across different environments.

paper_reviewer_data_quality_paper

Verdict: minor_revision
The manuscript is a survey chapter and does not present a primary dataset, so standard data quality checks for a new experimental study (e.g., missing value imputation strategies for a specific cohort) are not applicable. However, the review of data quality regarding **provenance, reproducibility of illustrative examples, and citation stability** reveals minor gaps.

First, several figures (specifically Figure 2, Figure 5, and Figure 6) explicitly state they are constructed using "simulated data" or "artificial" signals. While this is appropriate for a tutorial, the manuscript lacks a dedicated section or appendix detailing the **provenance of these synthetic datasets**. To ensure the illustrative examples are reproducible, the authors should cite the specific code repository or provide the parameters (e.g., noise levels, kernel widths, random seeds) used to generate the "artificial neuron" and "artificial LFP" traces. Without this, the "simulated" nature of the data in the figures is opaque, hindering the ability of readers to verify the visual claims.

Second, while the text references real-world datasets (e.g., in the caption of Figure 5, citing SedeEtal03, SedeEtal07a, etc.), it does not provide direct **links, DOIs, or repository paths** to the raw intracranial recordings. Given the sensitive nature of human iEEG data, the manuscript must clarify the **provenance and access restrictions** for these cited datasets. Readers should be directed to the specific institutional repositories (e.g., dbGaP, OpenNeuro) where these data can be accessed or verified, ensuring the data quality claims regarding "high-fidelity" recordings are grounded in accessible evidence.

Finally, the bibliography includes several entries marked as "In press" or solely as arXiv preprints (e.g., KumaEtal21, ScanEtal21, XieEtal21). To prevent **link rot** and ensure long-term data provenance, the authors should verify the final publication status of these works and update citations to include stable DOIs or volume/issue numbers where available. This is critical for a survey paper intended to serve as a long-term reference.

- [writing] The paper relies heavily on figures (e.g., Fig. 2, Fig. 5, Fig. 6) described as using 'simulated data' or 'artificial' signals to illustrate concepts. While acceptable for a tutorial, the manuscript must explicitly cite the specific synthetic data generation scripts or parameters used to ensure reproducibility of these illustrative examples. Without this, the 'simulated' nature of the data in the figures is opaque.
- [writing] The text references external datasets (e.g., 'data are from SedeEtal03, SedeEtal07a...') in Figure 5 captions without providing direct links, DOIs, or repository paths to the raw intracranial recordings. Given the sensitive nature of human iEEG data, the manuscript must clarify the provenance and access restrictions (e.g., via dbGaP or specific institutional repositories) for these cited datasets to allow verification of the data quality claims.
- [writing] The bibliography contains several entries with 'In press' or 'arXiv' status (e.g., KumaEtal21, ScanEtal21, XieEtal21). The manuscript should verify the final publication status and update citations to include stable DOIs or volume/issue numbers where available to prevent link rot and ensure long-term data provenance.

paper_reviewer_figure_critic

Verdict: minor_revision
The manuscript relies heavily on schematic figures to explain complex methodological concepts, which is appropriate for a survey chapter. However, several figures suffer from clarity and legibility issues that undermine their effectiveness.

Figure 1 (spatial_vs_temporal_resolution) is critical for establishing the paper's premise regarding intracranial recordings. The caption admits the axes are "not drawn to scale," which is a significant weakness. The visual overlap between the "gray shading" (experimental relevance) and the colored regions (modalities) is difficult to judge without a logarithmic grid or explicit tick marks. Readers cannot verify if the claimed "ideal" overlap for iEEG is accurate or an artifact of the drawing. The figure must be redrawn with a proper log-log scale to support the textual claims.

Figure 8 (superEEG) presents a severe technical issue. At ~33MB, the PDF is bloated, likely due to unoptimized vector paths or embedded images. This poses a risk for print legibility (lines may appear jagged or text may be too small) and digital accessibility. The correlation matrix in Panel D is particularly dense; the color map needs higher contrast, and the clustering boundaries should be thicker to remain visible when reduced to column width.

Figures 2 (signals) and 3 (patterns) use "arbitrary units" on the y-axes. While these are simulated data, the lack of physical units (e.g., millivolts, spikes/second) makes it harder for readers to map the schematic to real-world data. Adding a small inset scale bar or specifying units would improve the pedagogical value.

Finally, **Figure 5 (electrodes)** is a large file (~9MB). While the data density is high, the color coding for 53 patients in Panel B may be indistinguishable in grayscale print. Ensure the color palette is colorblind-safe and distinct enough to be differentiated without relying solely on hue, or provide a legend that groups patients by anatomical region rather than individual ID if the individual distinction is not critical.

- [writing] Figure 1 (spacetime): The caption explicitly states 'axes are not drawn to scale,' yet the figure is used to argue for specific spatiotemporal relevance ranges (gray shading). Without a scale or log-log grid lines, the relative positioning of modalities (e.g., iEEG vs. fMRI) is visually ambiguous and potentially misleading. Add a log-scale grid or a scale bar to validate the claimed overlaps.
- [writing] Figure 8 (superEEG): The file size (33MB) indicates excessive vector complexity or embedded high-res bitmaps. This will likely fail print legibility checks and cause rendering issues. Simplify the vector paths, reduce the number of data points plotted in the correlation matrices (Panel D), or convert to a high-DPI raster format (300– DPI) to ensure clarity at print scale.
- [writing] Figure 2 (signals) & Figure 3 (patterns): The y-axes in Panel A (spikes) and Panel B (LFP) are labeled 'arbitrary units.' For a methodological chapter, this reduces the figure's utility as a teaching tool. Specify the units (e.g., 'mV' for LFP, 'spikes/s' for firing rate) or provide a reference scale bar to ground the simulated data in physical reality.

paper_reviewer_jargon_police

Verdict: full_revision

The manuscript suffers from significant jargon overuse, frequently introducing complex technical terms, acronyms, and mathematical concepts without sufficient definition or plain-language explanation for a non-specialist audience.

First, acronyms are often used before being fully defined or are defined inconsistently. For instance, 'iEEG' and 'ECoG' appear in the abstract and early sections, but the full terms 'intracranial electroencephalography' and 'electrocorticography' are not always immediately clear or consistently expanded. Similarly, 'LFP' (Local Field Potentials), 'GLM' (Generalized Linear Models), 'MVPA' (Multivariate Pattern Analysis), 'RSA' (Representational Similarity Analysis), 'ISC' (Inter-subject Correlation), 'ISFC' (Inter-subject Functional Correlation), 'SRM' (Shared Response Model), 'HTFA' (Hierarchical Topographic Factor Analysis), 'TFA' (Topographic Factor Analysis), 'PCA' (Principal Components Analysis), 'ICA' (Independent Components Analysis), 'MEG' (Magnetoencephalography), 'fMRI' (functional

Magnetic Resonance Imaging), 'MRI' (Magnetic Resonance Imaging), and 'EEG' (Electroencephalography) are introduced with varying degrees of clarity. Many of these acronyms appear in the abstract and early sections without prior definition, violating standard scientific writing practices for accessibility.

Second, the text relies heavily on specialized jargon without adequate explanation. Terms like 'procrustean transformation', 'radial basis function', 'topographic factor analysis', 'hyperalignment', 'broadband shifts', 'phase coding', 'spike-triggered average', and 'searchlights' are used as if the reader is already familiar with them. For example, the section on 'Joint stimulus-activity models' introduces 'procrustean transformation' without a simple explanation of what it is (a geometric alignment method). Similarly, 'radial basis function' is mentioned in the context of TFA without a plain-language description of its role. 'Broadband shifts' and 'phase coding' are discussed in the context of neural signals without sufficient context for a general reader.

Third, the mathematical notation, while necessary for precision, is often presented without a preceding plain-language summary. The GLM section introduces the equation $y_t = f(x_t, \beta)$ and defines variables like N, M, and β in a dense manner. A brief sentence explaining the intuition behind the equation (e.g., "This equation models how neural activity predicts stimulus features") would greatly improve accessibility.

Finally, the text uses 'within-subject' and 'across-subject' interchangeably with 'within-participant' and 'across-participant', which can be confusing. Consistency in terminology is crucial for clarity.

To address these issues, the authors should: 1. Define every acronym at its first use, ensuring consistency throughout the text. 2. Provide immediate, plain-language definitions or explanations for all specialized jargon upon first introduction. 3. Precede complex mathematical equations with a brief, intuitive explanation of their purpose and meaning. 4. Standardize terminology (e.g., choose either 'subject' or 'participant' and stick with it). 5. Consider adding a glossary or a dedicated section explaining key terms for non-specialist readers.

These changes are essential to make the manuscript accessible to a broader audience, including researchers from adjacent fields and students who may not be experts in all the specific methodologies discussed.

- [science] Define every acronym at its first use, ensuring consistency throughout the text.
- [science] Provide immediate, plain-language definitions or explanations for all specialized jargon upon first introduction.
- [science] Precede complex mathematical equations with a brief, intuitive explanation of their purpose and meaning.
- [science] Standardize terminology (e.g., choose either 'subject' or 'participant' and stick with it).
- [science] Consider adding a glossary or a dedicated section explaining key terms for

non-specialist readers. These changes are essential to make the manuscript accessible to a broader audience, including researchers from adjacent fields and students who may not be experts in all the specific methodologies discussed.

paper_reviewer_logical_consistency

Verdict: minor_revision
The manuscript presents a comprehensive overview of methods for identifying stimulus-driven neural activity, but several logical inconsistencies and unsupported causal claims weaken the internal coherence of the argument.

First, in the section on **Joint stimulus-activity models** (Section 2.1.3), the author argues that neural responses are the "most direct representation" of stimulus properties because they are "uncorrupted by the experimentalists' assumptions." This conclusion directly contradicts the premises established in Section 1.1.2, where the text explicitly defines intracranial recordings (iEEG/ECoG) as aggregate measures of thousands of neurons, subject to volume conduction, and distinct from single-neuron firing rates. If the data is an aggregate, noisy proxy, it cannot logically be described as "uncorrupted." This contradiction creates a false dichotomy between "perfect" neural data and "imperfect" stimulus models, which undermines the logical justification for the proposed joint modeling approach.

Second, the transition from the problem of **variable electrode placement** to the necessity of **across-participant models** (Section 1.2) contains a logical gap. The text asserts that because electrode locations vary, "it can be difficult to identify reliable within-patient effects." This causal claim is unsupported. Within-patient analyses (such as GLMs or MVPA described in Section 2.1.1) rely on the temporal relationship between a specific patient's stimulus and their specific neural data; they do not require spatial alignment with other patients. The variability in electrode placement is a barrier to *across*-patient generalization, not a barrier to *within*-patient detection. The text conflates these two distinct analytical challenges, leading to a flawed justification for the exclusive focus on across-participant methods in certain contexts.

Finally, the logic underpinning **Inter-subject correlation (ISC)** (Section 2.2.1) relies on an unverified premise. The text claims that ISC isolates stimulus-driven activity because "non-stimulus-driven activity patterns are not expected to be correlated across people." This premise ignores the well-documented existence of correlated non-stimulus-driven signals, such as shared physiological artifacts (e.g., heart rate, respiration), common attentional fluctuations, or intrinsic network dynamics (e.g., default mode network fluctuations) that can be synchronized across subjects during naturalistic viewing. By failing to account for these confounds, the argument that ISC *specifically* identifies stimulus-driven patterns is logically incomplete. The conclusion that ISC is a "clean" measure of stimulus response does not strictly follow from the stated premises without additional assumptions about the independence of non-stimulus factors.

To resolve these issues, the author must clarify the limitations of neural data to avoid the "uncorrupted" claim, distinguish clearly between the requirements for within- vs.

across-patient analyses, and acknowledge the potential for non-stimulus confounds in ISC logic.

- [science] Claim that neural data is ”uncorrupted” (Sec 2.1.3) contradicts earlier admission of noise and aggregation (Sec 1.1.2).
- [science] Argument that variable electrode placement prevents within-patient analysis (Sec 1.2) is logically unsupported; within-patient GLMs do not require cross-subject alignment.
- [science] ISC logic assumes non-stimulus activity is never correlated across subjects (Sec 2.2.1), ignoring shared artifacts or attentional states that could cause false positives.

paper_reviewer_overreach

Verdict: minor_revision
The manuscript presents a broad survey of methods for identifying stimulus-driven neural activity, but it frequently over-claims the maturity and generalizability of the described techniques, particularly regarding multi-patient intracranial EEG (iEEG).

First, the abstract and introduction promise a review of specific approaches (GLMs, RSA, HTFA, etc.) illustrated by examples from recent literature. However, the body of the text largely functions as a conceptual tutorial on *what* these methods are, rather than a critical review of *how well* they work in the specific context of multi-patient iEEG. The paper lists these methods as if they are established, standard solutions, but fails to provide the comparative evidence or critical synthesis required to justify the claim that these are the definitive approaches for this specific, difficult problem. The ”examples” cited are often generic descriptions of the method’s logic rather than rigorous evaluations of their performance on the specific data constraints (sparse, non-overlapping electrodes) of the target domain.

Second, there is a significant over-claim regarding the robustness of ”across-participant” modeling in Section 2.4. Specifically, the description of Gaussian Process regression (SuperEEG) and Hyperalignment (Section 2.4.2) suggests these methods can reliably reconstruct full-brain activity or align functional spaces across patients with high fidelity. The text presents Figure 4 (SuperEEG) as a demonstration of reconstructing activity at unrecorded locations, but it glosses over the critical assumption that functional connectivity patterns are stationary and consistent across patients with different pathologies (e.g., drug-resistant epilepsy). In clinical populations, structural and functional abnormalities are common; assuming a ”global template” or correlation model derived from a heterogeneous group of patients accurately represents individual functional topographies is a strong claim that the paper does not sufficiently qualify. This risks misleading readers into believing these alignment techniques are more robust and generalizable than the current literature supports.

Finally, the concluding remarks over-extend the implications of the field’s current capabilities. The statement that these methods allow us to elucidate the neural basis of cognition "in ways that behavior alone cannot" is a strong claim. While true in principle, the paper does not adequately address the limitations of the "stimulus-driven" signal in iEEG, which is often weak, noisy, and heavily confounded by the specific clinical condition of the patients. The text implies a level of success in decoding and modeling that is not yet universally achieved, particularly for high-level cognitive processes in non-standardized electrode configurations. The review should temper these claims to reflect the current state of the art, acknowledging that while these methods are promising, they are not yet a panacea for the challenges of multi-patient iEEG analysis.

- [writing] The abstract claims to review specific approaches with examples, but the text functions as a high-level tutorial rather than a critical review. It over-claims its contribution by listing techniques without providing necessary critical synthesis, comparative performance data, or specific case studies justifying the depth implied by the title.
- [science] Sections 2.3.2 and 2.4.2 describe complex methods like Hyperalignment and SuperEEG as standard, robust solutions for multi-patient iEEG. The text glosses over severe limitations of applying these to sparse, non-overlapping grids in epilepsy patients, over-claiming the reliability of 'reconstructing' full-brain activity without qualifying assumptions about pathological brains.
- [writing] The conclusion over-extends by implying current iEEG methods have largely solved alignment and modeling problems. It fails to emphasize that for many high-level tasks, the 'stimulus-driven' signal remains weak and confounded by patient pathology, limiting direct extrapolation to general cognitive neuroscience.

paper_reviewer_safety_ethics

Verdict: minor_revision
This review focuses exclusively on safety and ethical considerations regarding the use of human intracranial data.

The manuscript correctly identifies the clinical context of the data source: patients with drug-resistant epilepsy undergoing invasive monitoring for seizure localization (Section 1.1.2, lines 230-240). This is a vulnerable population, and the ethical framework for their participation in research is critical. While the text notes that patients "may elect to participate in research studies," it lacks explicit confirmation of the regulatory safeguards in place. The manuscript must include a statement confirming that the research protocols were approved by an Institutional Review Board (IRB) or Ethics Committee and that informed consent was obtained from all participants (or their legal guardians) prior to data collection. It is essential to clarify that participation was voluntary and separate from their clinical treatment plan.

Furthermore, the discussion of "human-in-the-loop" techniques and automated modeling (Section 2.2.2) implies the processing of sensitive neural data. Given the unique identifiability of intracranial recordings and the small size of the patient cohorts often involved in such studies, the manuscript should briefly address data privacy and de-identification measures. Authors should confirm that data was anonymized or de-identified in accordance with relevant regulations (e.g., HIPAA, GDPR) before analysis or sharing.

Finally, the text states that invasive recordings are not appropriate for non-patient populations (Section 1.1.2, line 225). To prevent potential misinterpretation or misuse of the described methodologies, the manuscript should explicitly reiterate that these techniques are strictly limited to therapeutic clinical contexts and should not be adapted for non-therapeutic invasive research in healthy individuals. This clarification is vital for maintaining the safety boundary between clinical care and experimental neuroscience.

- [writing] The manuscript explicitly states that intracranial data comes from neurosurgical patients with drug-resistant epilepsy (Section 1.1.2, lines 230-240). It must explicitly confirm that all data usage was approved by an Institutional Review Board (IRB) and that informed consent was obtained from all participants, distinguishing between clinical care and research participation.
- [writing] The paper discusses 'human-in-the-loop' techniques and automated stimulus modeling (Section 2.2.2). It should briefly address data privacy and de-identification protocols used to protect patient identity, given the high sensitivity of intracranial neural data and the potential for re-identification in small, specific patient cohorts.
- [writing] The text mentions that invasive approaches are not appropriate for non-patient populations (Section 1.1.2, line 225). The review should explicitly state that the methods described are strictly limited to clinical populations and should not be interpreted as a protocol for non-therapeutic invasive research in healthy subjects.

paper_reviewer_scientific_evidence

Verdict: minor_revision
The manuscript is a comprehensive review chapter rather than a primary research article. As such, it does not present original experimental data, sample sizes, or statistical analyses that would allow for a direct evaluation of p-hacking risks, effect sizes, or replication within the text itself. The "scientific evidence" provided is entirely secondary, relying on the validity of the cited literature.

While the text effectively surveys methodological approaches (GLMs, MVPA, RSA, etc.), it occasionally presents the efficacy of these methods as a given without summarizing the quantitative strength of the evidence from the primary studies. For instance, in Section 3.2.1 regarding joint stimulus-activity models, the text describes the geometric intuition of aligning neural and stimulus trajectories but does not cite specific quantitative outcomes (e.g., cross-validated decoding accuracy or variance explained) from the referenced works (e.g., Haxby et al., 2011). To strengthen the evidentiary basis of the review, the author should briefly summarize the magnitude of effects or statistical robustness reported in the key primary studies for each method discussed.

Furthermore, in Section 3.3.2, the discussion of Inter-subject Correlation (ISC) and Inter-subject Functional Correlation (ISFC) omits a critical component of the scientific evidence:

the statistical controls used to validate these findings. ISC/ISFC analyses are prone to false positives if not properly controlled (e.g., via phase-shuffling surrogates or bootstrapping). The review should explicitly mention these standard validation procedures to demonstrate that the cited evidence for stimulus-driven activity is robust against alternative explanations like shared noise or low-level temporal autocorrelation.

Finally, the manuscript should include a clarifying statement in the introduction or conclusion explicitly distinguishing between the review's theoretical synthesis and the empirical evidence contained in the cited works, ensuring readers do not conflate the review's assertions with new statistical findings.

- [science] The manuscript is a theoretical review with no primary data, sample sizes, or statistical tests. It cannot be evaluated for p-hacking or effect sizes directly. The text must explicitly state that all evidence is secondary and no new empirical claims requiring statistical validation are made.
- [science] In Section 3.2.1, the text describes geometric alignment models but lacks quantitative metrics (e.g., variance explained, cross-validated accuracy) for cited examples. Summarize specific effect sizes or significance from primary literature to strengthen the evidence for these models' utility.
- [science] Section 3.3.2 discusses ISC/ISFC but omits statistical controls (e.g., phase-shuffling surrogates) needed to distinguish true stimulus-driven synchronization from spurious correlations. Briefly outline standard validation procedures to ensure cited evidence is robust against alternative explanations.

paper_reviewer_statistical_analysis

Verdict: minor_revision
The manuscript serves as a comprehensive methodological survey rather than a primary research article presenting new statistical findings. As such, there are no specific datasets, p-values, confidence intervals, or power analyses to evaluate for statistical appropriateness. The ”statistical analysis” lens is therefore limited to assessing the conceptual accuracy of the described methods and the completeness of the methodological discussion regarding statistical rigor.

The text provides a high-level overview of Generalized Linear Models (GLMs), Multivariate Pattern Analysis (MVPA), and Representational Similarity Analysis (RSA). However, in Section 3.1.2, when describing RSA and searchlight analyses, the manuscript fails to mention the critical statistical step of assessing significance. In practice, RSA searchlights involve thousands of correlation tests, necessitating rigorous multiple-comparisons correction (e.g., False Discovery Rate or cluster-based permutation tests). The absence of this discussion leaves a gap in the statistical guidance provided to readers.

Furthermore, in Sections 3.2.1 and 3.2.2, the descriptions of Hierarchical Topographic Factor Analysis (HTFA) and Gaussian Process (GP) regression are conceptually sound but lack a discussion of model assumptions. For instance, GP regression relies heavily on the choice of kernel function (e.g., radial basis function) and the assumption of stationarity in the spatial correlation structure. Similarly, HTFA relies on specific prior distributions. The review should briefly note these assumptions and the sensitivity of results to their violation, as this is crucial for the reproducibility and validity of applying these methods to intracranial data.

Finally, the manuscript should explicitly state in the abstract or introduction that it is a review of methods and does not contain primary statistical results. This distinction is vital to prevent readers from expecting empirical evidence, effect sizes, or statistical validation within the text itself.

- [writing] The manuscript is a conceptual review chapter and does not present primary statistical analysis, p-values, confidence intervals, or power calculations. Consequently, specific statistical assumptions and multiple-comparisons handling cannot be evaluated. The text should explicitly clarify that it is a methodological survey rather than an empirical study to avoid reader confusion regarding statistical rigor.
- [writing] In Section 3.1.2 (RSA), the text describes computing correlations between correlation matrices but omits discussion of statistical significance testing (e.g., permutation tests) or correction for multiple comparisons when performing searchlight analyses. As a review, it should cite standard practices for these corrections to guide future researchers.
- [writing] Section 3.2.1 (HTFA) and 3.2.2 (Gaussian Processes) describe complex hierarchical and spatial models. The review should briefly address the statistical assumptions inherent in these models (e.g., stationarity in GP kernels, prior distributions in HTFA) and the potential impact of violating them on the robustness of across-patient inferences.

paper_reviewer_text_formatting

Verdict: minor_revision
The manuscript demonstrates a generally high standard of LaTeX hygiene, with a clear hierarchy of sections and well-placed figures. However, several text formatting inconsistencies require attention before final submission.

First, there is a notable inconsistency in citation commands. The document loads ‘natbib‘ and predominantly uses ‘\citep‘ for parenthetical citations. However, on line 138, the command ‘\cite{Barr08}‘ is used instead of ‘\citep{Barr08}‘. This inconsistency should be resolved by standardizing all citations to ‘\citep‘ to maintain uniformity with the document class and package settings.

Second, a minor spacing error exists on line 138: ”...\citep{AzevEtal09, MotaHerc14} , which also...”. There is an unnecessary space between the closing brace of the citation and the following comma. This should be corrected to ”...\citep{AzevEtal09, MotaHerc14}, which also...” to adhere to standard typesetting rules.

Third, the formatting of multi-panel figure captions is slightly inconsistent. For instance,

Figure 2 (line 335) uses ”A–D. Neural signals” with an en-dash, while Figure 3 (line 428) uses ”A. Neuronal firing” with a period. While both are acceptable, consistency across all figures (e.g., Figures 4, 5, 6, 7, 8, 9) would improve the professional appearance of the document. Specifically, check the spacing after the panel labels (A, B, C) to ensure it is uniform.

Finally, a typo in the mathematical description on line 568 reads ”a the M-dimensional”. The word ”a” is redundant and should be removed. Additionally, while the equations are generally well-formatted, a quick review of all displayed math (e.g., lines 568, 604, 636) for consistent spacing around operators is recommended.

These issues are minor and easily fixable but contribute to the overall polish of the manuscript.

- [writing] Inconsistent citation command usage: The manuscript mixes \citep (natbib) and \cite (standard LaTeX) commands. For example, line 138 uses \cite{Barr08} while the rest of the document uses \citep. Standardize to \citep throughout to match the natbib package configuration.
- [writing] LaTeX spacing error: Line 138 contains an extraneous space before the citation command: '...\citep{AzevEtal09, MotaHerc14} , which also...'. Remove the space before the comma to ensure proper formatting.
- [writing] Figure caption formatting: Several figure captions (e.g., Fig. 2, line 335; Fig. 3, line 428) use bold text (\textbf) for panel labels (A, B, C) but the spacing between the label and the text is inconsistent. Ensure uniform spacing (e.g., 'A. Text' vs 'A–D. Text') across all multi-panel figure captions.
- [writing] Math mode hygiene: In the equation on line 568, the variable x_t is described as 'a the M-dimensional'. The word 'a' is a typo and should be removed. Additionally, ensure consistent spacing around mathematical operators in all displayed equations.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript is generally well-written, with a clear structure and logical flow that effectively guides the reader through complex methodological concepts. The abstract provides a concise overview, and the sectioning is logical. However, several minor typographical errors and grammatical slips detract from the overall polish and should be addressed before final publication.

Specifically, in Section 2.1 (”Building explicit stimulus models”), the word ”absense” is used instead of ”absence” (paragraph 2, line 3). Later in the same section, ”procede” appears where ”proceed” is intended (paragraph 4, line 12). In Section 1.3, there is a redundant repetition of the word ”participant” in the phrase ”single participant participant” (paragraph 3, line 10). The Summary section contains the misspelling ”drug-resistent” instead of ”drug-resistant” (paragraph 2, line 10). Additionally, in Section 3.2, the phrase ”taken taken” appears (paragraph 3, line 10).

The bibliography also contains a few errors: ”Americal” should be ”American” in the Spearman (2004) entry, and ”anlysis” should be ”analysis” in the Wang & Guo (2019) entry. Finally, ”interpet” is misspelled in Section 1.3 (paragraph 4, line 12).

While these errors are minor and do not obscure the scientific meaning, correcting them will significantly improve the professional quality of the text. The flow of arguments is strong, and the transitions between sections are smooth, but these surface-level issues should be cleaned up.

- [writing] Correct the typo ’absense’ to ’absence’ in Section 2.1 (Building explicit stimulus models), paragraph 2, line 3.
- [writing] Fix the typo ’procede’ to ’proceed’ in Section 2.1, paragraph 4, line 12.
- [writing] Remove the redundant word ’participant’ in Section 1.3, paragraph 3, line 10 (’...across repeated presentations to a single participant participant...’).
- [writing] Fix the typo ’drug-resistent’ to ’drug-resistant’ in the Summary section, paragraph 2, line 10.
- [writing] Correct the typo ’taken taken’ to ’taken’ in Section 3.2, paragraph 3, line 10.
- [writing] Fix the typo ’Americal’ to ’American’ in the bibliography entry for Spearman (2004).
- [writing] Correct the typo ’anlysis’ to ’analysis’ in the bibliography entry for Wang & Guo (2019).
- [writing] Fix the typo ’interpet’ to ’interpret’ in Section 1.3, paragraph 4, line 12.